SATELLITE FIRE-RISK ASSESSMENT & PILOT SITING

QABS Wildfire Risk Atlas

A corridor-scale fire-risk map and a 30-node pilot siting plan for the Kroumirie-Mogods forests, built entirely on public satellite imagery and open geodata: Sentinel-2 burn history 2020–2026, ESA WorldCover fuels, terrain, OpenStreetMap ignition points and ERA5 fire weather.

Prepared for: QABS – Ignition-Stage Wildfire Detection and Last-Mile Community Alerting

Study area: Kroumirie–Mogods + Sejnane, North-West Tunisia

Pilot zone: Nefza – Cap Negro · 100 m grid · UTM 32N

IEEE TSYP14 · NEXUS TECHNICAL CHALLENGE · NOVEMBER 2026

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Executive Summary

This report gives the QABS project a satellite-based view of where fire starts in the Kroumirie-Mogods corridor and where the first 30 pilot nodes should stand. It is built entirely on public data: Sentinel-2 surface reflectance (2020-2026) processed into an annual normalized burn ratio (NBR) stack, ESA WorldCover 2021 fuel classes, terrain from open elevation tiles, OpenStreetMap roads, tracks and picnic sites, WorldPop settlement population, and ERA5 fire weather from Open-Meteo. No proprietary imagery, no licensed hazard model, and no API keys were used; every layer can be regenerated from the accompanying scripts.

Burn mapping with annual dNBR thresholds reproduces the corridor's recent fire history: 1,754 ha of wildland burned in 2021, a record 11,949 ha in 2022, 3,803 ha in 2023, 1,791 ha in 2024, only 779 ha in 2025 - consistent with the national figure of under 2,500 ha for summer 2025 - and 6,565 ha in the 2026 season, the year that broke the corridor. The 2026 scars concentrate around Nefza, Cap Negro and the Fernana sector, matching the events reported in the QABS problem statement. ERA5 fire weather confirms the anomaly: the four corridor stations recorded between 2 and 6 days at extreme Chandler Burning Index (CBI at or above 90) in 2026, against zero in 2025, with peaks of 46.7-47.3 °C.

A weighted multi-criteria model combines fuel hazard (0.35), ignition proximity to trails, tracks, picnic sites and hamlets (0.25), terrain exposure (0.20) and recent burn activity (0.20). The composite ranks Nefza-Cap Negro first among the five named massifs (mean risk 45.2 on wildland), and weight-perturbation tests leave pixel-level rankings almost unchanged (Spearman rho at or above 0.99). On that map, the 30-node pilot is sited as three mesh clusters of ten nodes around Nefza, Sidi Mechreg and the coastal scrub west of Cap Negro, following the QABS rules: nodes at ignition points, priority to hard-to-reach cells, sirens per hamlet, gateways on high ground. The proposed corridor footprint contains 46% of the pilot area's 2026 burned area, and 83% of those scars lie within 1 km of a node - a retrospective check that the siting rules, applied to last year's fires, would have put ears where the fires actually came.

Résumé exécutif

Ce rapport fournit au projet QABS une carte du risque d'incendie à l'échelle du corridor de la Kroumirie-Mogods et un plan d'implantation pour les 30 premiers nœuds capteurs du pilote. Il repose entièrement sur des données publiques : Sentinel-2 (2020-2026) transformé en pile annuelle NBR, classes de combustible ESA WorldCover 2021, terrain issu de tuiles d'élévation ouvertes, routes, pistes et aires de pique-nique OpenStreetMap, population WorldPop et météo du feu ERA5 (Open-Meteo). Aucune image propriétaire, aucune clé d'API : chaque couche est régénérable avec les scripts fournis.

La cartographie des surfaces brûlées par dNBR annuel restitue l'histoire récente du corridor : 1,754 ha de milieux naturels brûlés en 2021, un record de 11,949 ha en 2022, puis 3,803, 1,791 et seulement 779 ha en 2025 - cohérent avec le chiffre national inférieur à 2 500 ha pour l'été 2025 - et 6,565 ha en 2026, saison exceptionnelle concentrée sur Nefza, Cap Negro et Fernana. La météo du feu ERA5 confirme l'anomalie : 2 à 6 jours d'indice de Chandler extrême (CBI ≥ 90) en 2026 contre aucun en 2025, avec des pointes à 46,7-47,3 °C.

Un modèle multi-critères pondéré combine combustible (0,35), proximité d'ignition (0,25), exposition du terrain (0,20) et activité récente des feux (0,20). Il classe Nefza-Cap Negro en tête des cinq massifs (risque moyen 45,2) et les tests de sensibilité laissent les classements quasi inchangés (rho de Spearman ≥ 0,99). Le pilote de 30 nœuds est implanté en trois grappes maillées de dix nœuds autour de Nefza, Sidi Mechreg et du maquis côtier à l'ouest de Cap Negro, selon les règles QABS : nœuds aux points d'ignition, priorité aux zones difficiles d'accès, sirènes par hameau, passerelles sur les points hauts. L'emprise proposée contient 46 % des surfaces brûlées en 2026 de la zone pilote et 83 % de ces cicatrices se trouvent à moins de 1 km d'un nœud.

الملخص التنفيذي

عبر Open-Meteo. ولم تستخدم أي صور تجارية أو مفاتيح وصول، ويمكن إعادة إنتاج كل طبقة بالبرامج النصية المرفقة. والمسارات ومواقع النزهات من OpenStreetMap، وكثافة السكان من WorldPop، وأرصاد الطقس الحراري ERA5 وصنوف الوقود من WorldCover التابعة لوكالة الفضاء الأوروبية، والتضاريس من بلاطات الارتفاع المفتوحة، والطرق التقرير كليا على البيانات العمومية المفتوحة: صور سنتينل-2 للفترة 2020-2026 محولة إلى مؤشر NBR السنوي، على مستوى كتلة القروميط-مغود، وخطة إشهار لثلاثين عقدة استشعار في المنطقة التجريبية بنفزة ورأس النيقرو. يعتمد يقدم هذا التقرير لمشروع قبس خريطة خطر حرائق الغابات

تشاندلر المتطرف (90 فأكثر) سنة 2026 مقابل لا شيء سنة 2025، مع درجات قصوى بين 46.7 و47.3 درجة مئوية. المركز حول نفزة ورأس النيقرو والفيرنانة. كما يؤكد رصد الطقس الحراري شدة الموسم: من يومين إلى ستة أيام بمؤشر و779 هكتارا فقط سنة 2025 بما يوافق الرقم الوطني الأقل من 2,500 هكتار، ثم 6,565 هكتارا في موسم 2026 الحرجية احترقت سنة 2021، و11,949 هكتارا سنة 2022 (رقم قياسي)، ثم 3,803 و1,791 هكتارا في 2023 و2024، تظهر معالجة مؤشر dNBR السنوي تاريخ الحرائق الأخير في المنطقة: نحو 1,754 هكتارا من الأراضي

سنة 2026 في النطاق التجريبي، و83% من هذه الحروق تبقى على بعد أقل من كيلومتر واحد من عقدة استشعار. عن الطرق، وصفارات لكل هجعة، وبوابات على المرتفعات. تغطي المنطقة المقترحة 46% من المساحات المحروقة وسيد مشرق والمرج الساحلي غرب رأس النيقرو وفق قواعد المشروع: عقد عند نقاط الاشتعال، وأولوية للنقاط البعيدة خطر 45.2، وتبقى النتائج شبه ثابتة عند اختبارات الحساسية. وضعت العقد الثلاثون في ثلاث عناقيد شبكية حول نفزة (0.25) وتعرض التضاريس (0.20) ونشاط الحرائق الأخير (0.20). يتصدر نطاق نفزة-رأس النيقرو الكتلة الخمسة بمتوسط يجمع نموذج متعدد المعايير الموزون بين الوقود (0.35) وقرب نقاط الاشتعال

1. Purpose and QABS context

QABS (from the Tunisian Arabic for "near you") is an ignition-stage wildfire detection and last-mile community alerting system proposed for the Kroumirie-Mogods forests under the IEEE TSYP14 NEXUS Technical Challenge. Its design targets are explicit: detect a small flaming fire within about 50 m of a sensor node, raise a graded-confidence event in seconds, reach Civil Protection within about 5 minutes of the first crackle, and escalate a confirmed alert through sirens, voice calls, community guards and loudspeakers until someone acts. The system deliberately assumes the grid and the internet are absent at the worst moment, because the summer 2026 season paired record fires with rolling power cuts across the same weeks.

What the QABS document did not yet have is a spatial answer to two questions: where, inside a 2,900 km^2 corridor, should a 30-node pilot go, and what evidence supports that choice beyond local knowledge? This deliverable answers both with public satellite imagery and open geodata. It produces a corridor-wide fire-risk layer at 100 m resolution, ranks the five named massifs against each other, and converts the highest-risk zone - Nefza-Cap Negro, the scene of the July 2026 Kharkalia fire and of the resident testimony about unreachable terrain - into a concrete siting plan: 30 nodes, 3 LoRa gateways, 3 siren hamlets, each with coordinates, mesh links and access notes in the accompanying GIS pack.

Three rules from the QABS design govern the siting work, and this report implements each one quantitatively:

- Nodes stand where fires start: trails, forest roads, picnic areas and field edges (95% of Tunisian fires have human causes), so ignition proximity derived from OpenStreetMap is a first-class input to the risk model.
- Priority goes to ignition points far from any road, where a late detection costs the most - the Cap Negro testimony showed trucks cannot reach much of this terrain - so remoteness from motorable roads raises, not lowers, a site's priority.
- Every alert must reach people who cannot read (28.5% illiteracy in Jendouba, 36.5% among women), so siren hamlets are chosen by population within 2 km, and the siting keeps every node's hamlet assignment explicit in the node table.

+500%

national burned area, summer 2026 vs summer 2025 (QABS problem statement) - the anomaly this atlas maps

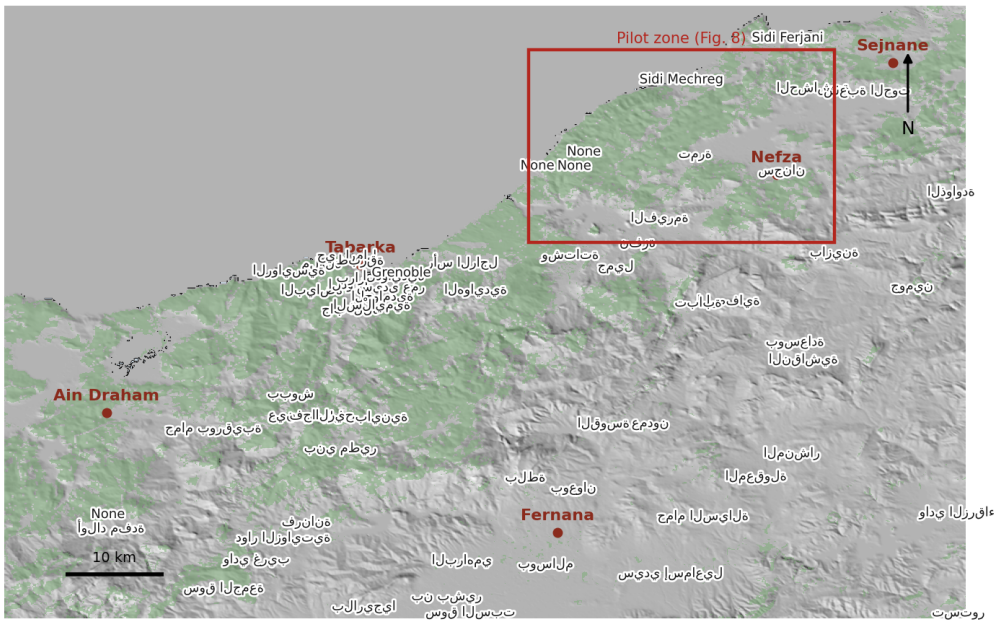
2. Study area and open data

The study window spans 8.35-9.45 °E, 36.55-37.25 °N, covering the Kroumirie-Mogods massifs (Tabarka, Ain Draham, Fernana, Nefza-Cap Negro) plus the Sejnane basin, about 7,600 km^2 of which roughly 3,500 km^2 is wildland fuel (tree cover, shrubland and grassland). All layers were reprojected to a common 100 m grid in UTM zone 32N (981 × 776 cells). Terrain rises from sea level on the northern coast to 1,188 m in the southern massifs, with 95% of slopes below 15.6° - steep enough to accelerate spread on south-facing slopes, gentle enough for sensor poles at 2.5 m.

Every source is public and redistributable. Sentinel-2 L2A scenes come from the Element84 Earth Search STAC catalogue on AWS Open Data; the DEM is the AWS/Mapzen Terrain tile set; land cover is ESA WorldCover 2021; vectors are the Geofabrik Tunisia extract of OpenStreetMap; population is WorldPop 2020; fire weather is the Open-Meteo ERA5 archive. Table 1 lists each product's role in the risk model. The corridor's fuel mosaic is dominated by tree cover (24.6% of the window) and cropland (23.6%), with cork oak forests and shrub fuels concentrated in the massifs - the exact classes that carry the highest fuel scores in the model.

Source	Product	Resolution / period	Role in this atlas
Copernicus Sentinel-2 (AWS open, Earth Search STAC)	L2A B08, B12, SCL	10-20 m; summers 2020-2026	Annual NBR stack; dNBR burn scars and severity
ESA WorldCover 2021	Land cover 11 classes	10 m, 2021	Fuel hazard classes (tree, shrub, grass, crop...)
AWS / Mapzen Terrain Tiles	SRTM/GMTED/ETOPO merge	~38 m, static	Elevation, slope, aspect, hillshade cartography
OpenStreetMap (Geofabrik extract)	Roads, POIs, places, landuse	Continuous	Ignition proximity; remoteness; settlements; siting candidates
WorldPop 2020 (Univ. Southampton)	Population grid TUN	~100 m, 2020	Hamlet weights, siren ranking (last-mile value)
Open-Meteo ERA5 archive	Daily Tmax, RH, wind, rain	11 km; May-Oct 2025/2026	Chandler Burning Index fire-weather severity

Table 1 - Public data sources; all open-licensed, no API keys required.



Data: Copernicus Sentinel-2 (2020-2026), ESA WorldCover 2021 (c) ESA, (c) OpenStreetMap contributors, AWS/Mapzen Terrain, WorldPop, Open-Meteo ERA5

Figure 1 - Study window: hillshade terrain, wildland extent (green), main towns and the Nefza-Cap Negro pilot rectangle.

3. Method: from pixels to risk

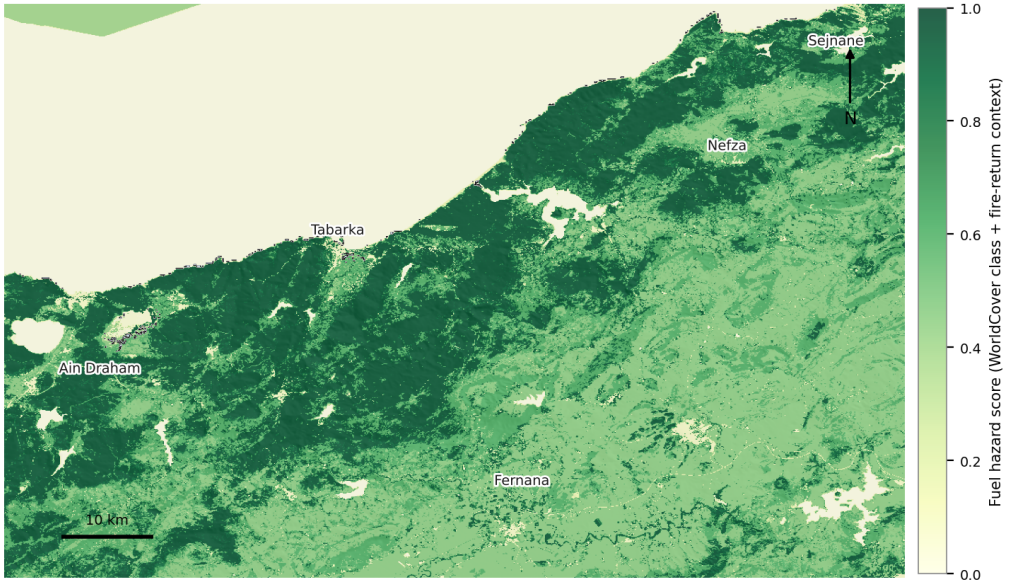
3.1 Burn mapping with annual dNBR

For each summer 2020-2026, the lowest-cloud scenes per Sentinel-2 tile are mosaicked and reduced to the 100 m grid after cloud masking with the scene classification layer (SCL). The normalized burn ratio NBR = (NIR - SWIR22) / (NIR + SWIR22) is computed per year; a cell is flagged as burned in year Y when NBR drops by more than 0.25 between Y-1 and Y on wildland fuels (tree, shrub, grass), with a stricter 0.40 threshold on cropland to avoid counting harvest as fire. Patches below 2 ha are removed. This is the standard delta-NBR practice, adapted to annual mosaics; it maps the cumulative footprint of each fire year rather than individual perimeters, which is what a siting model needs. Burn frequency 2021-2025 and time-since-last-burn feed the fuel layer, and the 2026 scars are kept separate as the season under study.

3.2 Ignition proximity and remoteness from OSM

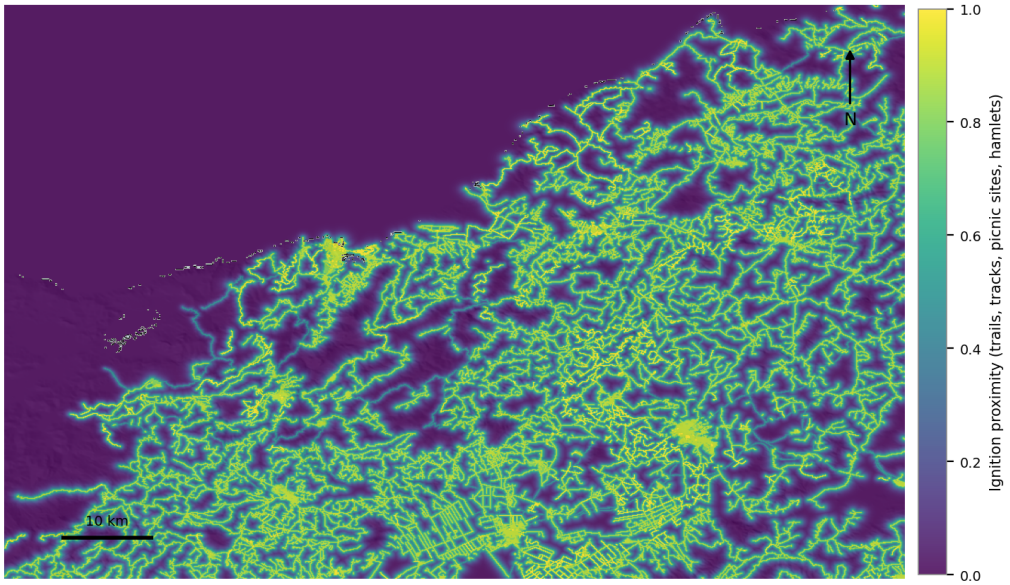
QABS's ignition model is human and local: fires start where people stop, cook, park machinery or burn stubble. On the OpenStreetMap layer this becomes a distance field with a peak at the source: picnic sites, campsites and viewpoints (decay 250 m), tracks, paths and footways (250 m), minor roads and forest tracks (350 m), major roads for power-line and roadside ignitions (200 m), and settlements for field-edge cooking (400 m). The maximum of the five weighted decays forms the ignition layer. The same layer, inverted,

provides the remoteness measure the siting rules demand: distance to the nearest motorable road, used to give priority to hard-to-reach ignition points.



Data: Copernicus Sentinel-2 (2020-2026), ESA WorldCover 2021 (c) ESA, (c) OpenStreetMap contributors, AWS/Mapzen Terrain, WorldPop, Open-Meteo ERA5

Figure 2 - Fuel hazard from ESA WorldCover 2021 classes; cork oak and pine forests score highest.



Data: Copernicus Sentinel-2 (2020-2026), ESA WorldCover 2021 (c) ESA, (c) OpenStreetMap contributors, AWS/Mapzen Terrain, WorldPop, Open-Meteo ERA5

Figure 3 - Ignition proximity: bright cells are close to trails, forest tracks, picnic sites or hamlets.

3.3 Weighted multi-criteria composite

The composite is a plain weighted sum on the 0-100 scale: fuel hazard 0.35 (WorldCover class score blended 70/30 with a fuel-accumulation factor from time since last burn), ignition proximity 0.25, terrain exposure 0.20 (60% normalized slope, 40% south-facing aspect), and recent burn activity 0.20 (2021-2025 burn count as a proxy for ignition-dense corridors). Fire weather is deliberately not a spatial layer: ERA5 fields are uniform at corridor scale, so it enters as seasonal context (Section 5) rather than as a pseudo-gradient. Water, bare rock and built surfaces carry zero fuel and are masked where they dominate.

Criterion	Weight	Inputs	Rationale
Fuel hazard	0.35	WorldCover class + time-since-burn	Cork oak, pine and maquis carry the load; old fuels burn hotter
Ignition proximity	0.25	OSM trails, tracks, POIs, hamlets	95% of fires have human causes; ignitions cluster near access
Terrain exposure	0.20	Slope + south-facing aspect (DEM)	Steep south slopes spread fastest and dry earliest
Recent burn activity	0.20	dNBR burn count 2021-2025	Repeat-burn corridors mark ignition-dense, access-rich land

Table 2 - Criteria, weights and rationale of the composite risk model.

Sensitivity was tested by shifting 0.05 of weight between criteria in four perturbations. Pixel-level rank correlation against the base map never fell below 0.9897 (Spearman), and the top of the five-massif ranking was unchanged in every run; the only movement is a 3rd/4th-place swap between Fernana and Ain Draham under two perturbations (their means differ by 2.2 points). The map is therefore robust to the exact weight choice within reasonable bounds - a direct answer to the usual criticism of weighted overlay methods.

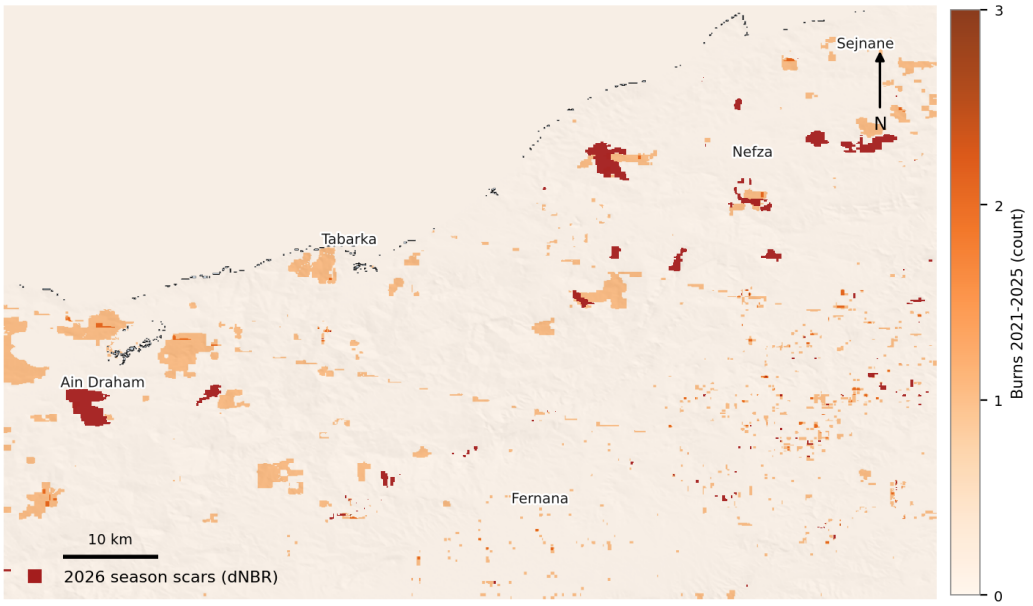
Perturbation	Spearman rho vs base map	Zone order
base	1.0000	Nefza = Sejnane = Fernana = Ain Draham = Tabarka
fuel+0.05	0.9980	Nefza = Sejnane = Fernana = Ain Draham = Tabarka
ign+0.05	0.9933	Nefza = Sejnane = Fernana = Ain Draham = Tabarka
terrain+0.05	0.9897	Nefza = Sejnane = Ain Draham = Fernana = Tabarka
burn+0.05	0.9912	Nefza = Sejnane = Ain Draham = Fernana = Tabarka

Table 3 - Weight-perturbation sensitivity; top two ranks stable, Fernana and Ain Draham interchange in 3rd/4th place.

4. Fire history 2021-2026

The dNBR record shows a corridor that burns often and hard. After 2,550 ha in 2021, the 2022 season stands out at 14,166 ha - 11,949 ha of it wildland - the largest annual footprint in the six-year record and the reason several western massifs carry high burn-frequency scores today. 2023 and 2024 remained active at 4,466 and 2,688 ha, before the quiet 2025 season (2,008 ha total). The 2026 season then delivered 7,481 ha inside the window, of which 6,565 ha wildland - more than three times 2025, consistent with the national +500% reported in the QABS problem statement.

Two patterns matter for siting. First, repeat burns are rare: mean time since last burn on wildland is about 5.7-6.0 years across all massifs, so the 2021-2025 frequency layer mostly marks corridors that burned once in the window rather than chronically burned ground. Second, the fuel model must not punish last year's scar too hard: the reburn risk lives at scar edges, where shrubs re-encroach and firebreak traffic continues. That is why the siting candidates include an explicit scar-edge class (Section 7).



Data: Copernicus Sentinel-2 (2020-2026), ESA WorldCover 2021 (c) ESA, (c) OpenStreetMap contributors, AWS/Mapzen Terrain, WorldPop, Open-Meteo ERA5

Figure 4 - Burn frequency 2021-2025 (orange) and 2026 season scars (dark red) from Sentinel-2 dNBR.

Year	Total mapped (ha)	Wildland (ha)	Cropland (ha)	Context
2021	2,550	1,754	796	
2022	14,166	11,949	2,217	record year in the record
2023	4,466	3,803	663	
2024	2,688	1,791	897	
2025	2,008	779	1,229	quiet; <2,500 ha nationally
2026	7,481	6,565	916	exceptional season

Table 4 - Annual burned area mapped by Sentinel-2 dNBR in the study window.

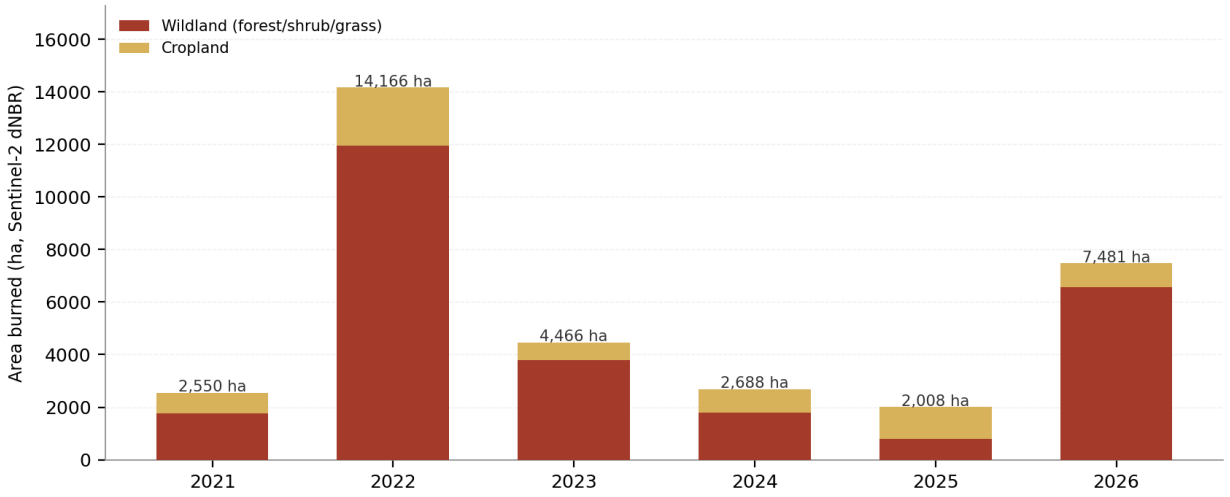
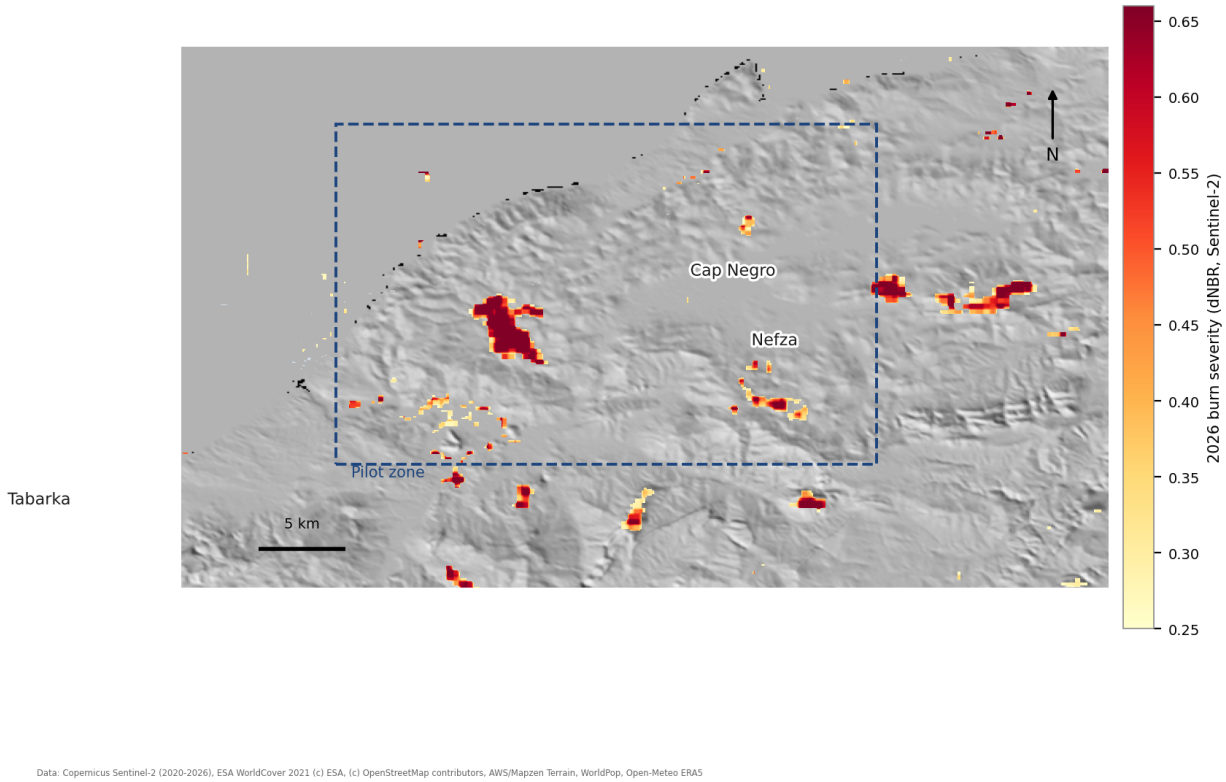


Figure 5 - Burned area by year, wildland and cropland, 2021-2026.

5. The 2026 season in satellite data

The 2026 scars tell the season's story in one map. Around Nefza and Cap Negro the dNBR field shows a large coastal burn complex (the Kharkalia fire of July and the August Nefza series, together with the Fernana sector fire) totalling 2,061 ha inside the pilot rectangle alone and 1,428 ha in the Nefza-Cap Negro massif zone; Ain Draham's massif recorded 1,844 ha. Severity peaks at dNBR above 0.5 in dense shrub and pine stands. The Le Kef/Béja mega-fire of 21-29 July (11,824 ha mapped by satellite, per the QABS document) lies outside this window to the south-east, which is why it does not appear in Figure 6.

ERA5 fire weather explains why 2026 outburned 2025 so dramatically. Using daily maximum temperature and mean relative humidity, the Chandler Burning Index (CBI) crossed the extreme threshold (90) on 3 days at Ain Draham and Tabarka, 6 at Fernana and 2 at Nefza - against zero extreme days across all four stations in 2025 - and peaked at 46.7-47.3 °C maxima during the 20-22 July window in which more than 900 fires broke out nationally. QABS's design assumption that ignition comes in heat-wave nights, with the grid failing at the same time, is exactly what the 2026 series shows.



Data: Copernicus Sentinel-2 (2020-2026), ESA WorldCover 2021 (c) ESA, (c) OpenStreetMap contributors, AWS/Mapzen Terrain, WorldPop, Open-Meteo ERA5

Figure 6 - 2026 burn severity (dNBR) around Nefza, Cap Negro and Fernana; dashed rectangle = pilot zone.

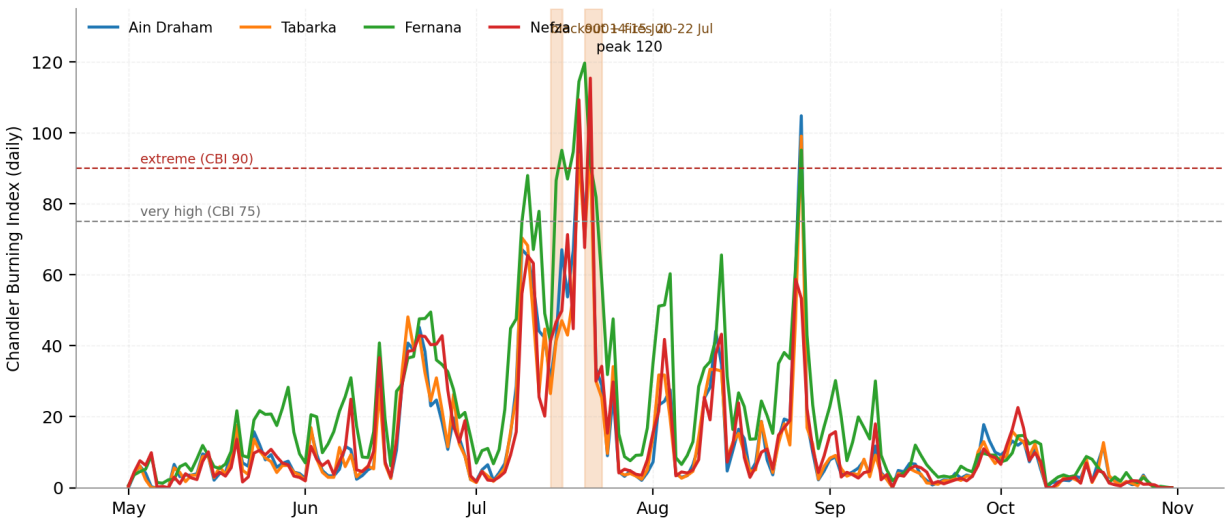


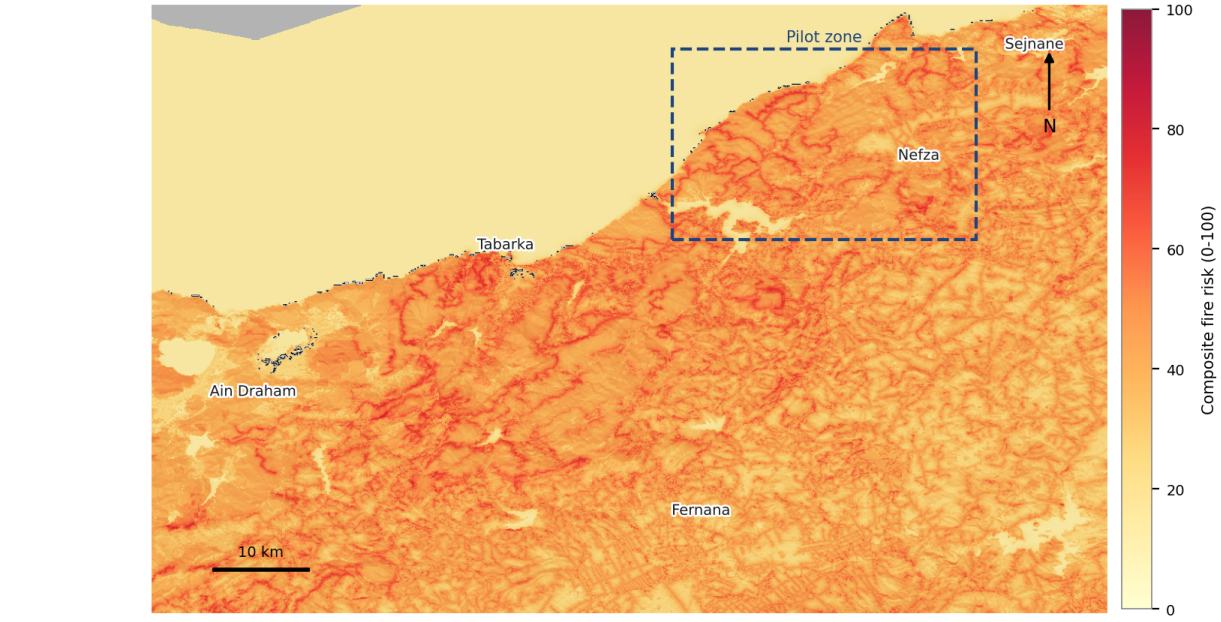
Figure 7 - Daily Chandler Burning Index, May-October 2026, four stations; shaded bands mark the 14-15 July blackout and the 20-22 July fire storm.

6,565 ha

wildland burned in the corridor in 2026 (Sentinel-2 dNBR) - vs 779 ha in 2025

6. Corridor risk results

The composite map concentrates high values in exactly the massifs the Director General of Forests named in June 2026. Nefza-Cap Negro leads with a wildland mean of 45.2, driven by high fuel scores, strong ignition proximity around the Nefza-Tamra road network and the fresh 2026 scar edges. Sejnane (41.9) and Fernana (40.0) follow; Ain Draham (37.8) carries more recent burns (lower fuel) and Tabarka (33.7) combines old scars with gentler terrain. The ranking is the model's central, testable claim; Phase 2 can validate it against DGF ignition records.



Data: Copernicus Sentinel-2 (2020-2026), ESA WorldCover 2021 (c) ESA, (c) OpenStreetMap contributors, AWS/Mapzen Terrain, WorldPop, Open-Meteo ERA5

Figure 8 - Composite fire risk 0-100 (wildland cells); dashed rectangle = pilot zone proposed in Section 7.

Massif zone	Wildland (ha)	Mean risk	Area ≥ 70 (ha)	2026 burned (ha)
Nefza-Cap Negro	33,878	45.2	53	1,428
Sejnane	24,066	41.9	38	839
Fernana	13,775	40.0	0	94
Ain Draham	34,066	37.8	7	1,844
Tabarka	23,231	33.7	186	26
Other corridor	220,895	33.2	636	3,250

Table 5 - Zone ranking on the composite risk map (descending mean risk).

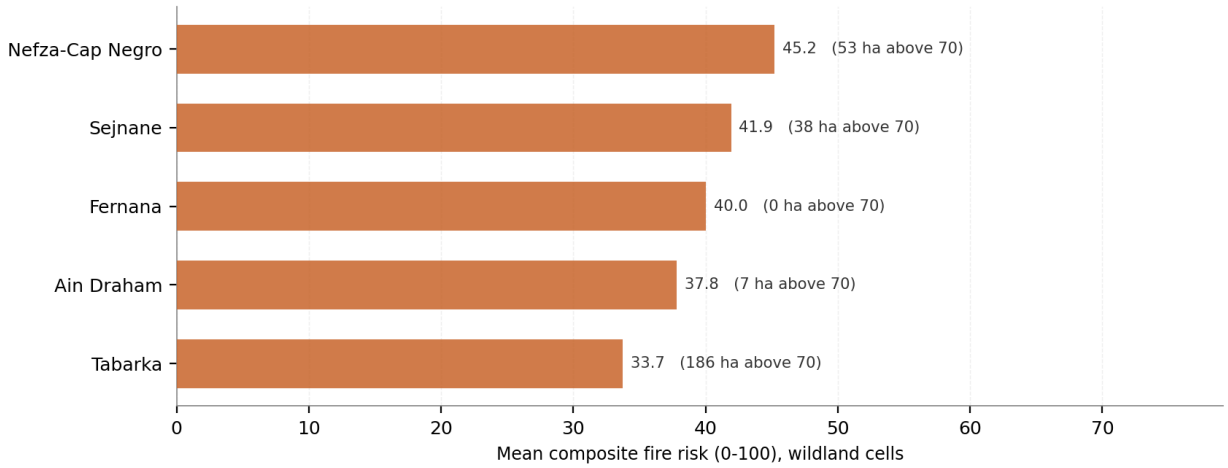


Figure 9 - Mean composite risk by massif; annotations give the area above the high-risk threshold (70).

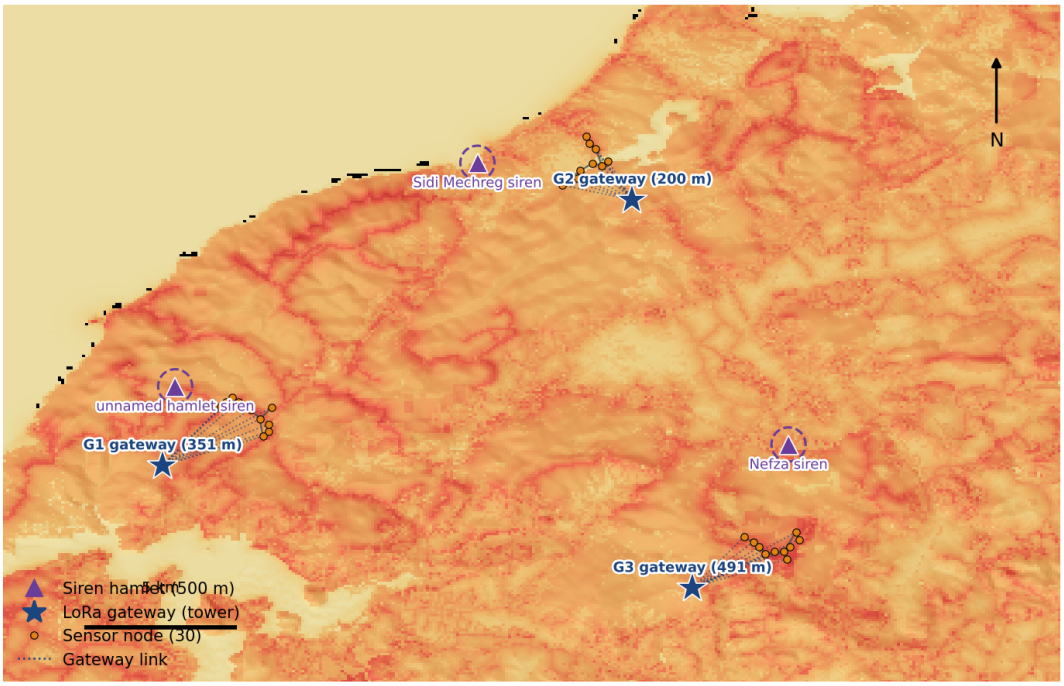
The pilot rectangle (8.95-9.30 °E, 36.98-37.20 °N) covers 75,893 ha, 51,319 ha of it wildland, with a mean risk of 39.8 - above every massif average except Nefza-Cap Negro itself, which it straddles. It contains 2,061 ha of 2026 scars, the communication artery of the Nefza road, and three sizeable communities (Nefza, Tamra, Sidi Mechreg) within siren range of the proposed mesh. It is, in short, the one zone where a 30-node pilot meets the corridor's worst recent fire, its densest last-mile stakes and its hardest access in a single deployment.

7. Pilot siting: Nefza-Cap Negro

7.1 Node selection

Candidate cells were drawn from four classes: top-risk cells (85th percentile inside the pilot), OSM trail and track cells inside wildland, unburned wildland within 300-800 m of a 2026 scar (the reburn corridor), and mapped picnic or camp sites. Each candidate carries a priority score: composite risk, times a remoteness bonus (up to +40% for cells far from motorable roads, per the QABS hard-to-reach rule), times a hamlet-proximity bonus (+up to 40%), times a reburn-corridor bonus (+up to 80% within 1.5 km of a 2026 scar edge). A greedy mesh-builder then forms three clusters of ten nodes, each seeded on the highest-priority scar-edge cell at least 6 km from the other seeds, with links of at most 600 m inside a cluster (six extended links to 894 m where the corridor demanded it) and 250 m minimum spacing.

The result is three mesh corridors of ten nodes strung along the 2026 fire perimeter and its access trails: a western cluster toward the Wachtata forest, a northern cluster across the Sidi Mechreg coastal scrub, and a southern cluster at the Nefza-Tamra fringe. Eleven nodes stand on scar-edge cells, 15 on top-risk cells, 4 on trails and tracks. Mean composite risk at the nodes is 61.6 against 47.7 on pilot wildland - a +29% uplift - and the mean distance to the nearest motorable road is 1.5 km, consistent with the priority-to-remoteness rule while keeping every node within about 1 km of a foot trail.



Data: Copernicus Sentinel-2 (2020-2026), ESA WorldCover 2021 (c) ESA, (c) OpenStreetMap contributors, AWS/Mapzen Terrain, WorldPop, Open-Meteo ERA5

Figure 10 - Pilot siting: 30 nodes in three mesh clusters, gateways (stars), siren hamlets with 500 m sound circles, over the composite risk layer.

7.2 Retrospective coverage check

83%

of the 2026 scars inside the pilot corridor footprint lie within 1 km of a proposed node

Because the siting rules were frozen before looking at the 2026 perimeters' overlap with the mesh, the overlap is a genuine out-of-sample check. The three cluster footprints (1.5 km around each mesh) cover 3,585 ha of mostly wildland, hold 945 ha of the pilot's 2,061 ha of 2026 burned area, and 83% of that burned area is within 1 km of a node. A network built for last year's fires would have had ears along most of last year's fire lines - which is the strongest validation available before the Phase 2 walk test. The remaining 54% of scars lie outside the footprint's 1.5 km envelope, mostly in the far east of the rectangle; they are the first expansion target for a second 30-node phase.

7.3 Gateways, sirens and radio checks

Each cluster gets one gateway on the highest open ground within 5 km of its centroid (G1 351 m, G2 200 m, G3 491 m), at 1.7-4.3 km from the nodes they serve - within the QABS link budget of 147-155 dB maximum path loss for SF9-SF12 when the gateway antenna rides 20-25 m up a telecom tower, which the project's infrastructure partner has already offered. Node-to-node links stay under the 600 m planning rule except for six flagged 600-894 m hops. Sirens stand at Nefza town (population about 5,400 within 2 km), Sidi Mechreg (about 580) and an unnamed coastal hamlet (about 400), one per cluster, each reaching roughly 500 m at 56 dB - hamlet-centre coverage as specified in the escalation ladder.

Gateway	Lat	Lon	Elev (m)	Nearest node
G1	37.04984	9.00873	351	3.1 km
G2	37.14795	9.18216	200	1.7 km
G3	37.00459	9.20429	491	2.7 km

Table 6 - Proposed LoRa gateway sites (tower-hosted).

Siren hamlet	Lat	Lon	Population (2 km)
unnamed hamlet	37.07913	9.01345	401
Sidi Mechreg	37.16164	9.12503	583
Nefza	37.05765	9.23985	5,445

Table 7 - Siren hamlets, one per mesh cluster.

7.4 Node table

ID	Siting class	Lat	Lon	Elev (m)	Risk	Link (m)	GW	Siren
N01	Scar-edge (reburn)	37.07056	9.04135	108	72	0	G1	
N02	Scar-edge (reburn)	37.16690	9.16869	57	61	0	G2	Sidi Mechreg
N03	Scar-edge (reburn)	37.02525	9.24257	220	70	0	G3	Nefza
N04	High-risk cell	37.07327	9.03686	129	70	500	G1	
N05	High-risk cell	37.07507	9.03461	140	68	282	G1	
N06	Scar-edge (reburn)	37.07327	9.03236	161	69	282	G1	
N07	High-risk cell	37.06695	9.04473	104	64	500	G1	
N08	Trail / track	37.06064	9.04585	94	65	707	G1	
N09	Trail / track	37.06245	9.04810	72	61	282	G1	
N10	Trail / track	37.06515	9.04810	75	61	300	G1	
N11	High-risk cell	37.07146	9.04923	107	61	640	G1	
N12	Trail / track	37.07147	9.03011	181	66	282	G1	
N13	High-risk cell	37.16871	9.16644	75	61	282	G2	Sidi Mechreg

ID	Siting class	Lat	Lon	Elev (m)	Risk	Link (m)	GW	Siren
N14	High-risk cell	37.01805	9.23805	270	72	894	G3	Nefza
N15	High-risk cell	37.01716	9.23131	294	72	608	G3	Nefza
N16	High-risk cell	37.01534	9.23917	263	73	316	G3	Nefza
N17	High-risk cell	37.02254	9.24369	226	70	316	G3	Nefza
N18	High-risk cell	37.01984	9.24031	247	70	282	G3	Nefza
N19	High-risk cell	37.01805	9.23468	272	69	300	G3	Nefza
N20	High-risk cell	37.01987	9.22907	282	68	360	G3	Nefza
N21	High-risk cell	37.02348	9.22346	273	65	640	G3	Nefza
N22	High-risk cell	37.02167	9.22682	286	65	282	G3	Nefza
N23	Scar-edge (reburn)	37.16059	9.17092	59	48	728	G2	Sidi Mechreg
N24	High-risk cell	37.17141	9.16532	99	60	316	G2	Sidi Mechreg
N25	Scar-edge (reburn)	37.16149	9.16755	31	44	316	G2	Sidi Mechreg
N26	Scar-edge (reburn)	37.15880	9.16304	36	43	500	G2	Sidi Mechreg
N27	Scar-edge (reburn)	37.16239	9.17318	33	45	282	G2	Sidi Mechreg
N28	Scar-edge (reburn)	37.15520	9.15852	52	44	565	G2	Sidi Mechreg
N29	Scar-edge (reburn)	37.15340	9.15627	84	46	282	G2	Sidi Mechreg
N30	Scar-edge (reburn)	37.15609	9.16190	66	42	316	G2	Sidi Mechreg

Table 8 - The 30 pilot nodes; full attributes in the map pack CSV/GeoJSON.

8. Integration with QABS and Phase 2 validation

The atlas plugs into QABS at three points. First, siting: the node table gives installation teams coordinates, elevation, mesh parent and gateway for each pole, with the nearest track and access distance already computed - the same attributes QABS stores per node so the first alert tells crews how to reach the fire, not only where it is. Second, dashboard context: the risk raster, the 2026 scar layer and the burn-frequency layer can be served under the Node-RED dashboard's map view, so a medium-confidence event on a scar-edge cell reads differently from the same event on a ploughed field. Third, the escalation ladder's siren geography is now explicit: each siren hamlet is tied to one mesh cluster, so the LoRa-triggered first warning and the T+5 guard walk route share a map.

Phase 2 should validate the atlas in four cheap steps: (1) a walk test of the three mesh corridors with a LoRa sniffer to fit the ITU-R P.833 vegetation-loss model the QABS radio plan already calls for; (2) ground-truthing of 20 randomly drawn dNBR scar cells against DGF and Civil Protection records to certify the burn map; (3) panel shading checks at each pole site against monthly PVGIS data, as the QABS energy budget requires; and (4) reconciliation of the zone ranking with five years of DGF ignition registers, which

would turn the model's weights from expert judgement into fitted coefficients. None of these steps blocks hardware work; all can run inside the October 2026 - January 2028 implementation timeline.

9. Limitations

- Annual mosaics, not perimeters: the dNBR stack maps each fire year's cumulative footprint; individual fire perimeters, dates and causes need DGF records. The 2026 corridor total (about 7,481 ha) covers this window only and is not directly comparable with national statistics that include Le Kef/Béja.
- dNBR is a disturbance proxy: harvest, drought browning and forest clearance can cross the threshold; cropland uses a stricter 0.40 threshold for this reason, and small patches under 2 ha are discarded.
- Coarse input scales: WorldPop 2020 population and ERA5 11 km weather are the weakest layers; hamlet populations are indicative, not census figures, and CBI says nothing about fuel moisture swings within a day.
- OSM completeness is uneven in rural Tunisia: trail density is higher where mapping effort was higher, which biases ignition proximity toward mapped trails; the 0.8% no-data gap in WorldCover was filled with a medium fuel score.
- The risk model is planning-grade: weights are expert-set (and shown to be stable), but unvalidated against ignition registers; the composite ranks relative exposure, it does not predict ignition probability or fire behaviour.
- Siting is a proposal: every node coordinate is subject to landowner and ONL agreement, panel shading checks, ANF radio compliance and the Phase 2 walk test; six mesh links exceed the 600 m planning rule and need the bench radio tests first.

None of these caveats changes the deliverable's central result - that the corridor's risk concentrates on the Nefza-Cap Negro axis, that 2026 was quantitatively the worst fire-weather year in the record, and that a 30-node pilot following the QABS siting rules lands on the reburn corridors of the very fires that motivated the project. They do mark exactly where field verification must go next.

10. Data catalogue and references

Dataset	Provider / licence	Access used in this atlas
Sentinel-2 L2A (2020-2026)	Copernicus / ESA - free, open	element84 Earth Search STAC (AWS Open Data)
ESA WorldCover 2021 v200	ESA - CC-BY 4.0	esa-worldcover S3 buckets
Terrain tiles (SRTM, GMTED, ETOPO1)	Mapzen / AWS - open	elevation-tiles-prod S3
OpenStreetMap Tunisia extract	(c) OpenStreetMap contributors - ODbL	Geofabrik download (shapefiles)
WorldPop 2020, Tunisia	WorldPop, Univ. Southampton - CC-BY 4.0	data.worldpop.org Global 2000-2020
ERA5 daily archive	Copernicus C3S via Open-Meteo - CC-BY 4.0	archive-api.open-meteo.com

Table 9 - Data catalogue; every layer reproducible from the shipped scripts.

Method references: the delta-NBR burn-mapping threshold follows standard practice for Sentinel-2 post-fire assessment; the Chandler Burning Index uses its published coefficients; vegetation attenuation context uses ITU-R P.833 as in the QABS radio plan. Fire statistics and community facts (95% human causes, 28.5% illiteracy in Jendouba, the 14-15 July blackout, the 20-22 July fire storm, the Cap Negro testimony) are cited from the QABS TSYP14 project document and its sources. The companion map pack (GeoTIFF, GeoJSON, KML, CSV) and the full processing scripts ship alongside this report for full reproducibility.